AI-Assisted Proposal Triage Architecture & Process Flow

Reference: MATACS-TH-2026-02 | End-to-End Workflow: Ingestion, Hybrid AI/Rule Triage, Human Sign-Off

STAGE 1: INGESTION & CONFIG

1.1 PDF Proposal Upload

Reviewer uploads bids (A-E)
Text & OCR conversion

1.2 Admin Dynamic Config

Dynamic criteria & weights
Budget CAD 260k | 16-wk ceiling

1.3 Data Isolation & Tagging

Wrap text in <proposal_data>
Strict Data vs Instruction boundary

Isolated Data

STAGE 2: AI EXTRACTION & SAFETY

2.1 Adversarial Injection Scan

LLM Guardrail detects directives
(e.g., Proposal D pre-approval trick)

Clean / Flagged

2.2 Structured LLM Extraction

temp=0.0 | Pydantic JSON Schema
Extracts specs, licence, terms, costs

Raw JSON

2.3 Source Quote Verifier

Regex matches extracted values
against verbatim text in source PDF

Verified Facts

STAGE 3: RULES & SCORING

3.1 Mandatory Pass/Fail Gate

Rule: Bidder itself holds licence?
Rule: WSIB & 5M insurance valid?
(Proposal B fails subcontractor)

Passed

3.2 Ceiling & Completeness Gate

Price <= CAD 260k | Time <= 16w
(Proposal D flagged: no schedule)

Complete

3.3 Deterministic Scoring Formula

Multi-attribute formula (0-100):
Exp 30%, Price 25%, Time 15%,
Warranty 15%, Team 15%

Triage State

Flagged / Excluded

Final Scores

STAGE 4: HUMAN AUDIT & DECISION

4.1 Triage Review Dashboard

Ranked: A (98.8), E (80.5), C (77.8)
Ineligible: B | Flagged: D
Evidence quotes highlighted

4.2 Human Overrides & Sign-Off

Reviewer validates disqualification,
resolves flags, or adjusts score
(Mandatory written rationale)

4.3 Tamper-Evident Audit Log

Immutable log: doc hashes, prompt,
model version, raw extraction, scores,
reviewer edits & timestamps

4.4 Final Award Decision

Official procurement officer
makes binding legal award

Architecture Responsibilities & Design Principles:

Ingestion & Config:	Admin dynamic criteria screens + document ingestion with strict data/instruction isolation tags.
AI Layer (Probabilistic):	LLM entity extraction (temp=0.0) + adversarial injection detection + source citation quotes.
Rules / Code (Deterministic):	Mandatory pass/fail regex, hard ceiling enforcement, and zero-hallucination math scoring.
Human Oversight (Review):	Reviewer triage dashboard with one-click verification of ineligibility and manual override capabilities.
Audit & Final Award:	Cryptographically verifiable audit log tracking all edits, prompt versions, and human award decision.